CSE221 Lesson Plan, Spring 2026
Mid-term:
Final:

Course Outline: https://docs.google.com/document/d/1l0ub-Dn-8wnYDPwRU2ce8Nrtal

Class #	Topics	Materials	Reading Tasks	Problem Solving Tasks
0	Ice breaking	https://m.media-amazon.com/images/I/51oXcGxCoFL._AC_SX679_.jpg		
1	1. Introduction: Algorithms for accuracy, efficiency (space, time); Algorithms in real world 2. Time Complexity (recap): RAM Model, Loop iterations vs execution time, Growth Functions, best-worst scenario	Iterative time complexity: https://drive.google.com/file/d/1rTgD7Vzsvd4jWFr9taPqrdttOHlHXR	RAM Model : https://drive.google.com/file/d/104buTIJIA8dgrAidKkvZHV_tNV Characterizing Running Times : https://drive.google.com/file/d/1Z-TxrsHxjpfN	Past Questions Iterative Time Complexity (Sheet I) : https://www.comp.nus.edu.sg/~cs1020/tut/15s2/tut09ans/T9_ans.p Iterative Time Complexity (Sheet II) : https://www.cs.princeton.edu/courses/archive/spring20/cos226/pre
2	Searching algorithms 1. Binary: Logical explanation on the logarithmic time complexity, fitness/feedback function, finding upper - lower bound 2. Ternary: Concept, Advantages over binary search 3. Balancing between no. of search steps and cost of each step: why quaternary search is not that popular	Linear: https://drive.google.com/file/d/1vAJ2EbjdNmv6_ZgbEcDwDJj3RnwPs7qp/view Binary: https://drive.google.com/file/d/1P6QjoYShah1YIFzdgNCRbnXd379BiuQb/view Ternary: https://drive.google.com/file/d/1tP_ebyE3uHpGto6NIcDf5-VIzUsgoyOe/view		
3	Sorting algorithms 1. Case analysis on N^2 sorting algorithms: Bubble, Insertion/Selection; # of iterations, # of swaps on best/worst scenario 2. Linear (case specific) time sorting: Count sort, a good example of why nested iterations is not always multiplied for time complexity	Animated simulations from: https://visualgo.net/en/sorting	Insertion Sort : https://drive.google.com/file/d/1IaJcGbReNU71cjAL-jJKAE18p CountSort : https://drive.google.com/file/d/13hy-QbTtl-23yV04Yiunqpoe77OC	
4	Sorting contd. 1. NlgN sorting: Merge, Quick (fixed and random pivots)	Merge: https://drive.google.com/file/d/1JUGTeiVhRnwaMKVcvwruklKGymlq-U25/view?usp=s Quick: https://drive.google.com/file/d/1vbcbOtf5jkflJBLzMbniFr3FZIvF3nbO/view?usp=sharin	Merge Sort : https://drive.google.com/file/d/1L1W_tBDOjdPtWUV8Zx5Spyjnx Quick Sort Book Chapter : https://drive.google.com/file/d/1zho4ekYmu3m-IIM Quick Sort Book Chapter (Alternative) : https://drive.google.com/file/d/1KYTy5	
5	Quiz + Lagged discussion (if any)			
6	Asymptotic Notations: upper, lower, tight bound complexity	Iterative time complexity: https://drive.google.com/file/d/1rTgD7Vzsvd4jWFr9taPqrdttOHlHXR		
7	Divide and Conquer (DnC) algorithms 1. Look back to merge, quick sort 2. Max-sum subarray: O(NlgN) [can also be solved in O(N^3), O(N^2), O(N)]	Max-sum subarray: https://docs.google.com/presentation/d/1rIC6OXh7W5ELsxkfGbnYWJRS		
8	DnC contd.: Karatsuba multiplication	Karatsuba: https://docs.google.com/presentation/d/1xcTFIl_wxMCVomD54YsKQxytc6VfNaf3	Karatsuba (Dasgupta) : https://drive.google.com/file/d/1isJG1yX4dohpBMtCn Karatsuba (Errickson) : https://drive.google.com/file/d/1VSK3qK44gxP9_2XQ	
9	Time complexity of recursive/DnC algorithms: recursion tree, substitution method, master theorem	Recursive Time complexity: https://drive.google.com/file/d/12cR_JBR8wZg0NDAIl-wzL6fUsZ	Master Theorem (Notes) : https://docs.google.com/document/d/1AvPn-0KIBf5	Master Theorem (Problems) : https://www.csd.uwo.ca/~mmorenom/CS424/Ressources/master.pdf
10	BFS: traversal (review), shortest path on unweighted graph, Bicolorable/Bipartite graph detection, cycle detection (odd or even length)	bfs dfs: https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWgk6V91q/view?usp=		
11	Quiz + Review			
Mid Semester Examination				
12	DFS: traversal (review), Edge classification, cycle detection	bfs dfs: https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWgk6V91q/view?usp=		
13	What is DAG, Topological ordering, Strongly Connected Components (SCC, Kosaraju's algo)	top sort: https://drive.google.com/file/d/1U-PVynHjD-PytCeHvcqtCsCaEE-iL-qc/view?usp=sha SCC: https://drive.google.com/file/d/1fckjavRyW4yCAi2zmAjsF3pKhIsr3FJb/view?usp=sharin		
14	Shortest path algo: What is SSSP and APSP, Dijkstra's algo, ~~*Bellman-Ford Algo* (Negative-weight edges in play)~~	MST+Dijkstra: https://drive.google.com/file/d/1zNpTudRm--5wPHKrownXyosZQ1ta69F3/view		
15	MST: Kruskal's (+DSU)	MST+Dijkstra: https://drive.google.com/file/d/1zNpTudRm--5wPHKrownXyosZQ1ta69F3/view		
16	Quiz + Lagged discussion (if any)			
17	Greedy: Huffman coding, ~~*Interval scheduling (covered in lab before mid)*~~	Greedy Basics, time scheduling: https://drive.google.com/file/d/1U-LLjWQHP-RdARYV8vOYl Huffman: https://drive.google.com/file/d/1-_oOWBKQARdaWgZvLTUFlV5tCYd5CqZ4/view?u		
18	DP-1: Fractional vs 0-1 Knapsack	fractional knapsack: https://drive.google.com/file/d/1U-LLjWQHP-RdARYV8vOYlMkKglZBFzc DP Basics, 0-1 knapsack: https://drive.google.com/file/d/1tdh2T7G4L4uMVuLnxvqHhJqVxku		
19	DP-2: LCS, Recurrence relation, Properties: overlapping subproblem and optimal substructure	LCS: https://drive.google.com/file/d/1YJonhR6glZoGMLk0G6323MJytohIIvn6/view?usp=shar		
20	Quiz + Review			
Final Examination				

Marks Distribution	Quiz: 20 (best 3 out of 4, or as instructed by course teacher) Assignment: 05 (4 assignments) Mid Semester Exam: 20 Final Exam: 30 Lab: 25